

be a major problem when adopting the IoT on a larger scale, and a lot depends on what the user already has in hand. Over a period of time, even without us realising, we are already upgrading the infrastructure. These are invisible upgrades. An example is the wireless network that started with no wireless communication and moved to 1G, 2G and 3G, and then to 4G LTE and WiMAX.

In some cases, it requires partial or complete replacement of hardware. These are visible upgrades, like sensors or Wi-Fi-enabled bulbs. Many systems and protocols these days make it very easy to add an IoT component to the existing device. For example, systems based on highway addressable remote transducer (HART) protocol can be easily upgraded to WirelessHART, by using an adapter and a gateway. "But, sometimes the cost or effort required to backward-integrate can be so high that it would be advisable to replace the existing equipment with a new system altogether," says Nihal Kashinath of IoTBLR.

### Is IoT security a catastrophe waiting to happen

Any communication that happens over-the-air needs to be secure. But, functionality and safety are found to be interdependent. Increasing the functionality will dilute the safety, as entrepreneurs focus their time, effort and resources on developing better features. In a similar way, giving more focus on security hampers functionality. The challenge is to strike the right balance between these aspects, and this becomes the deciding factor for long-term success of any system.

Nihal Kashinath says that current IoT systems are not very secure because there is a lack of seriousness about security, given that attacks on connected devices have been sporadic. This is dangerous, since security should be in the fab-

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ric of the connected-devices system, not as a separate module that is tacked on later.

However, T. Anand believes that, as of today, systems are secure enough. In the long run, the design and architecture of these systems or applications would tell more. Cases of network security attacks are very few, considering the number of connected devices available these days, observes Shibi Salim. Developers are becoming more vigilant and focusing well on security aspects.

"When it comes to data and bearer security, it is all managed by very well-defined standards and protocols," says Amit Shah. This is a multi-layered approach. Security of data that we are sending on the cloud has multiple layers—social security number (SSN), username and password—and, a lot of sensitive data is obfuscated or made anonymous, so that even if you get hold of it, you cannot make sense of it.

Then comes security of access—who can get hold of the data. When access is secured, data is segregated and controlled by a unique identifier, so one person cannot get hold of another person's data.

Many people are too intimidated by the idea of companies or people getting access to data that they might not want to share with them. Some examples are the positioning data Google receives through Maps app or the use of phones to take photos in high-security regions, thus losing privacy in the digital world.

To solve such issues, there are solutions like geo-fencing that allow you to switch-off your device completely, switch to flight mode or disable certain functions when inside a virtually-defined location, thus protecting real-world privacy.

### From inflated expectations to the plateau of productivity

Gartner's hype cycle report might have the IoT as its most over-hyped technology in 2014, but their analysts believe that it will only take five to ten years for it to reach maturity, or the plateau of productivity, according to the hype cycle. Coincidentally, the prediction for Big Data reaching the same level of maturity is also five to ten years. The year 2025 sure looks set to be an extremely promising year for us technologists. ●